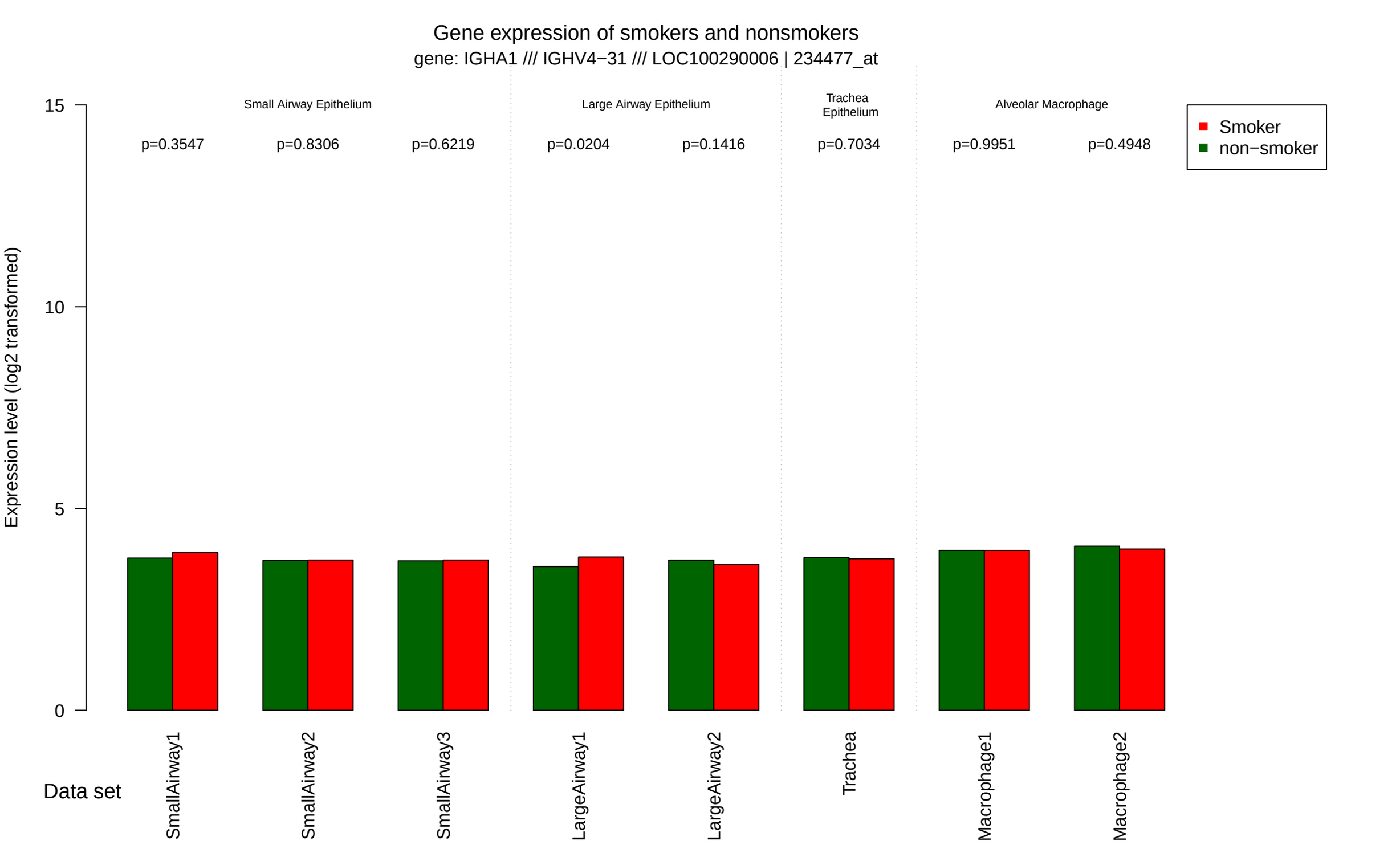
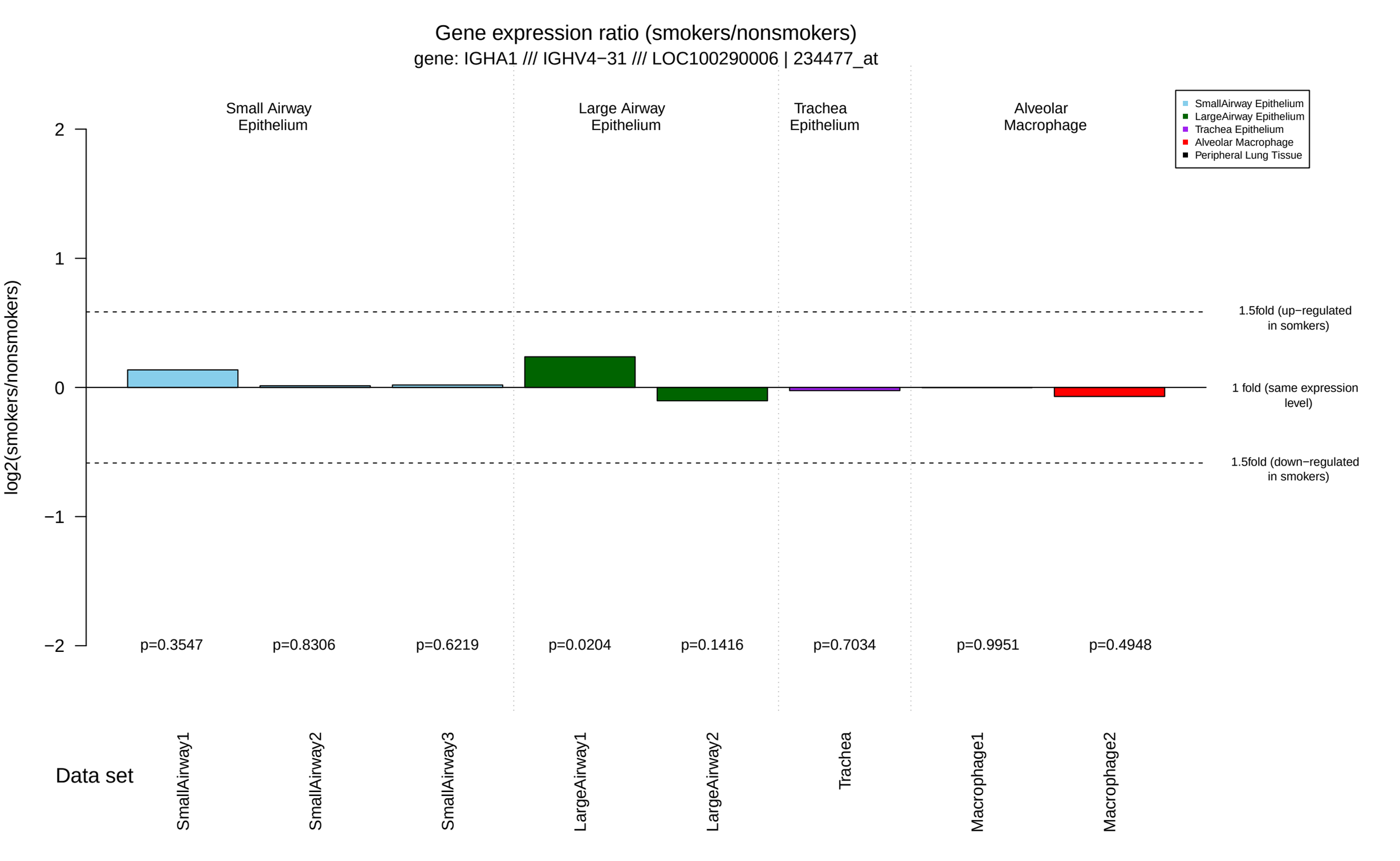
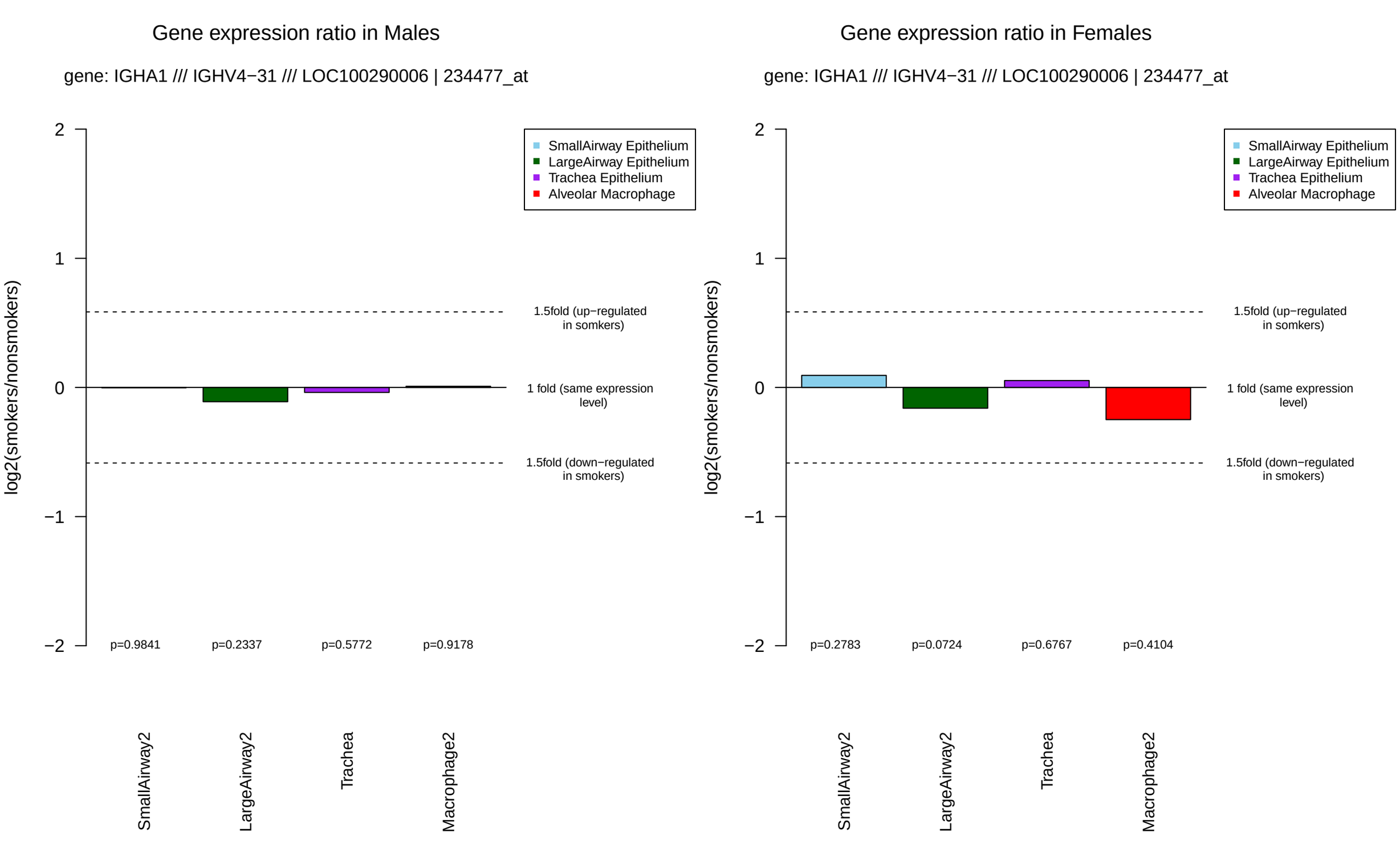
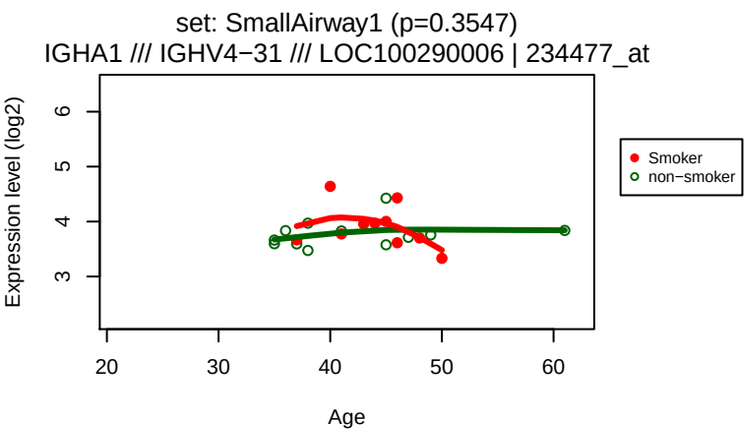
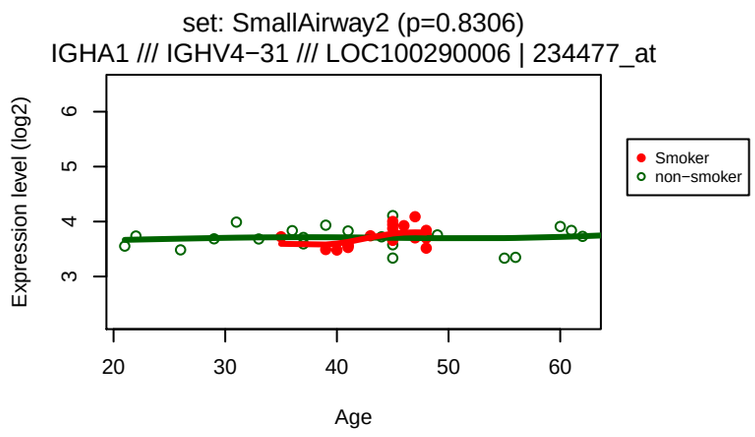
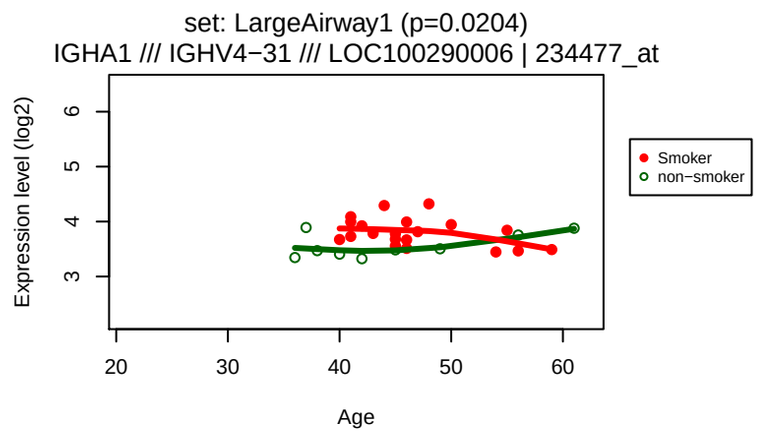

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHV4−31 /// LOC100290006 | 234477_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3547
p=0.8306
p=0.6219
p=0.0204
p=0.1416
p=0.7034
p=0.9951
p=0.4948
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

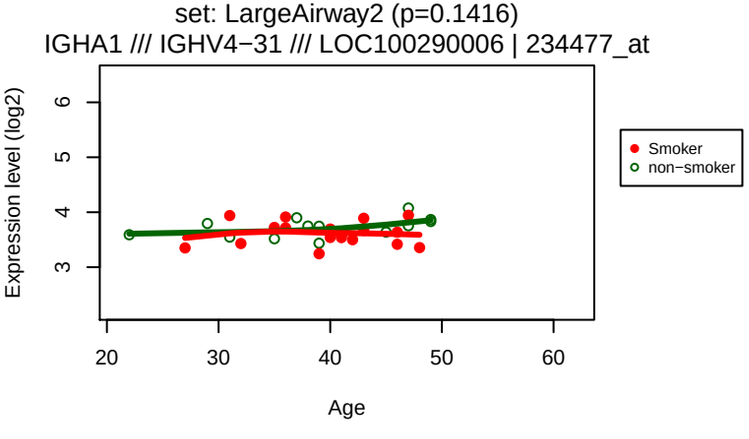

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHV4−31 /// LOC100290006 | 234477_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3547
p=0.8306
p=0.6219
p=0.0204
p=0.1416
p=0.7034
p=0.9951
p=0.4948
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

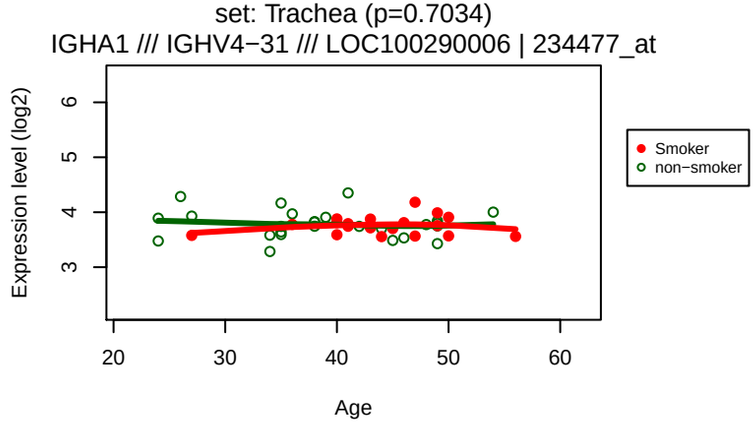

Gene expression ratio in Males
gene: IGHA1 /// IGHV4−31 /// LOC100290006 | 234477_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9841
p=0.2337
p=0.5772
p=0.9178
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHA1 /// IGHV4−31 /// LOC100290006 | 234477_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2783
p=0.0724
p=0.6767
p=0.4104
SmallAirway2
LargeAirway2
Trachea
Macrophage2

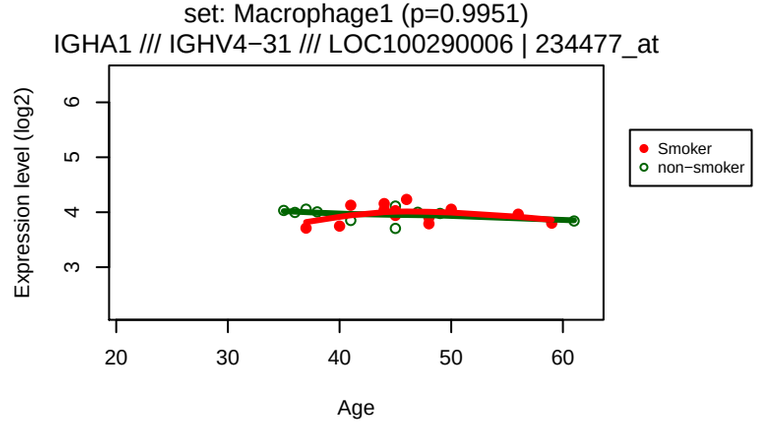

set: SmallAirway1 (p=0.3547)
IGHA1 /// IGHV4−31 /// LOC100290006 | 234477_at
Expression level (log2)
Age
Smoker
non−smoker

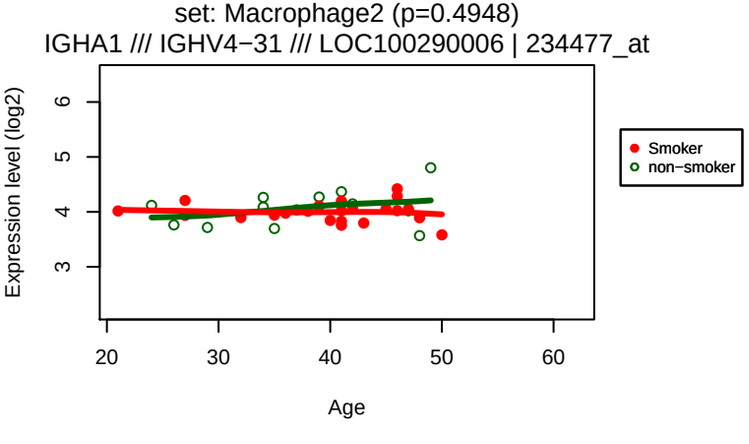

set: SmallAirway2 (p=0.8306)
IGHA1 /// IGHV4−31 /// LOC100290006 | 234477_at
Expression level (log2)
Age
Smoker
non−smoker

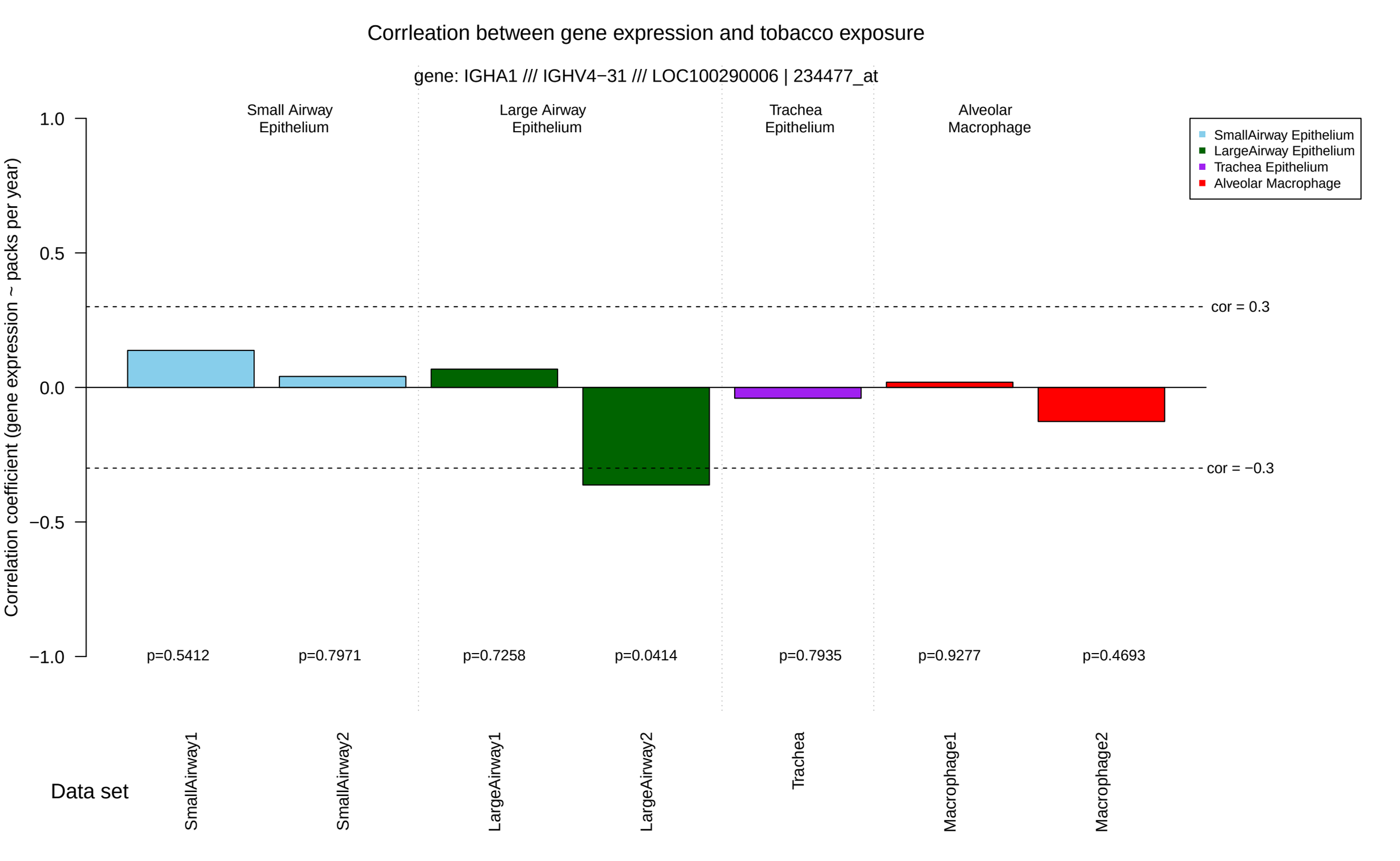

set: LargeAirway1 (p=0.0204)
IGHA1 /// IGHV4−31 /// LOC100290006 | 234477_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1416)
IGHA1 /// IGHV4−31 /// LOC100290006 | 234477_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7034)
IGHA1 /// IGHV4−31 /// LOC100290006 | 234477_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9951)
IGHA1 /// IGHV4−31 /// LOC100290006 | 234477_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4948)
IGHA1 /// IGHV4−31 /// LOC100290006 | 234477_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHV4−31 /// LOC100290006 | 234477_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5412
p=0.7971
p=0.7258
p=0.0414
p=0.7935
p=0.9277
p=0.4693
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2